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# Linear Independent Component Analysis via Optimal Transport Metric as Contrast

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## Abstract

Independent Component Analysis (ICA) recovers mutually independent sources from linear mixtures. Classical algorithms—FastICA, JADE, and InfoMax—compute non-Gaussianity via surrogate contrast functions (negentropy approximations, fourth-order cumulants, parametric log-likelihoods) that fail on heterogeneous source distributions. We propose OT-ICA, which uses the 1D squared Wasserstein distance ( $W_2^2$ ) to a standard Gaussian as the contrast function, computed exactly by quantile sorting without density estimation. We prove  $W_2^2$  satisfies a weighted mixture upper bound with strict inequality for non-Gaussian sources, establishing it as a valid ICA contrast. Across  $d \in \{10, 20, 30\}$  and five mixture regimes ( $N = 10,000$ , 10 trials), OT-ICA achieves lower Amari error than FastICA, JADE, InfoMax, and Picard on all continuous and heterogeneous distributions. On purely discrete sources the contrast is present, but step-function CDFs produce gradient plateaus that stall the solver. Two applications are presented: ocular artifact removal from frontal EEG, and structural shock identification in Vector Error Correction Models for financial price discovery.

## 1 INTRODUCTION

Independent Component Analysis (ICA) [Jutten and Herault, 1985, Comon, 1994] recovers mutually independent source signals  $\mathbf{S}$  from a linear mixture  $\mathbf{X} = \mathbf{A}\mathbf{S}$ . Identifiability holds under the following conditions [Comon, 1994].

**Theorem 1** (Linear ICA Identifiability). *Let  $\mathbf{X} \in \mathbb{R}^d$  be observed data and  $\mathbf{S} \in \mathbb{R}^d$  latent sources. Assume (i)  $\mathbf{X} = \mathbf{A}\mathbf{S}$  with  $\mathbf{A}$  invertible, (ii)  $s_1, \dots, s_d$  mutually independent, and (iii) at most one  $s_i$  Gaussian. Then  $\mathbf{A}$  is identifiable up*

*to column permutation and scaling.*

The Central Limit Theorem implies that linear mixtures of independent non-Gaussian sources are more Gaussian than their constituents. ICA recovers sources by maximizing marginal non-Gaussianity over the orthogonal group [Hyvärinen et al., 2001], justified by the following identity.

**Theorem 2** (Independence and Non-Gaussianity [Cardoso, 2022]). *Let  $Y$  be whitened ( $\text{Cov}(Y) = \mathbf{I}$ ). The mutual information satisfies:*

$$\mathcal{J}(Y) = G(Y) - \sum_i G(Y_i) \quad (1)$$

where  $G(\cdot)$  denotes KL divergence to the best-fitting Gaussian. Since  $G(Y)$  is invariant under orthogonal rotation, minimizing  $\mathcal{J}(Y)$  is equivalent to maximizing  $\sum_i G(Y_i)$ .

(Proof via KL Pythagorean identities on the Product and Gaussian manifolds: Appendix A.)

Evaluating  $G(Y_i)$  exactly requires the marginal density. Classical algorithms replace it with surrogates: FastICA [Hyvärinen et al., 2001, Amari et al., 1996] uses logcosh negentropy approximations, JADE [Cardoso and Souloumiac, 1993] fourth-order cumulants, InfoMax [Bell and Sejnowski, 1995] a parametric log-likelihood, and Picard [Ablin et al., 2018] an adaptive nonlinearity with a provably converging solver. Each proxy fails on specific source geometries (Appendix E).

We propose OT-ICA, using  $W_2^2$  to a standard Gaussian as the contrast. The Monge–Kantorovich problem [Monge, 1781, Kantorovich, 1942] defines the  $p$ -Wasserstein distance between measures  $\mu$  and  $\nu$  as:

$$W_p(\mu, \nu) = \left( \inf_{\pi \in \Pi(\mu, \nu)} \int \|x - y\|^p d\pi(x, y) \right)^{1/p} \quad (2)$$

In 1D, Brenier’s theorem [Brenier, 1991] and Caffarelli’s regularity [Caffarelli, 1992] guarantee that the optimal map

is monotone, reducing  $W_2^2$  to:

$$W_2^2(\mu, \nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^2 dt \quad (3)$$

This is computable exactly by quantile sorting. Prior work uses 1D  $W_2$  for normality testing [del Barrio et al., 1999] and distribution comparison [Peyré and Cuturi, 2019]; we formalize  $W_2^2$ -to-Gaussian as a valid ICA contrast with finite-sample convergence guarantees.

## 2 THE OT-ICA FRAMEWORK

After centering and whitening  $\mathbf{X}$  to  $\tilde{\mathbf{X}}$ , the unmixing matrix is constrained to the Orthogonal Group  $\mathcal{S}$  [Cardoso, 2022]. OT-ICA finds the projection  $\mathbf{b}$  maximizing  $W_2^2$  to the standard Gaussian  $\Gamma \equiv \mathcal{N}(0, 1)$ :

$$\hat{\mathbf{b}} = \arg \max_{\mathbf{b} \in \mathcal{S}} W_2^2(\mathbb{P}_{\mathbf{b}^\top \tilde{\mathbf{X}}}, \Gamma). \quad (4)$$

The objective is maximized at a pure source direction, established by two bounds.

**Theorem 3** ( $W_2$  Upper Bound). *Let  $\mathbf{Z} = (Z_1, \dots, Z_d)^\top$  be a vector of mutually independent latent sources, and let  $\mathbf{b}$  be a weight vector on the unit sphere ( $\sum_{i=1}^d b_i^2 = 1$ ). The squared Wasserstein distance between the linear mixture  $\mathbf{b} \cdot \mathbf{Z}$  and the standard normal distribution  $\Gamma$  is bounded by the weighted sum of the source distances to  $\Gamma$ :*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (5)$$

**Theorem 4** (Strict  $W_2$  Inequality for Non-Gaussian Sources). *Assuming at most one source  $Z_i$  is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3 is a strict inequality for any valid mixture where multiple weights  $b_i$  are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (6)$$

The global maximum is attained when  $\mathbf{b}$  aligns with a single source direction. Proofs via comonotonicity contradictions under Caffarelli’s regularity theory are in Appendix B.

### 2.1 METHODOLOGICAL ENHANCEMENTS

Three components are required for optimization on  $\mathcal{S}$ : (i) **Analytical Gaussian Targets** [Fournier and Guillin, 2015], replacing sampled quantile targets with their closed-form bin expectations to eliminate approximation noise; (ii) **Riemannian Gradient Retraction** [Absil et al., 2008],

Baseline Validation — Amari Error: 0.015

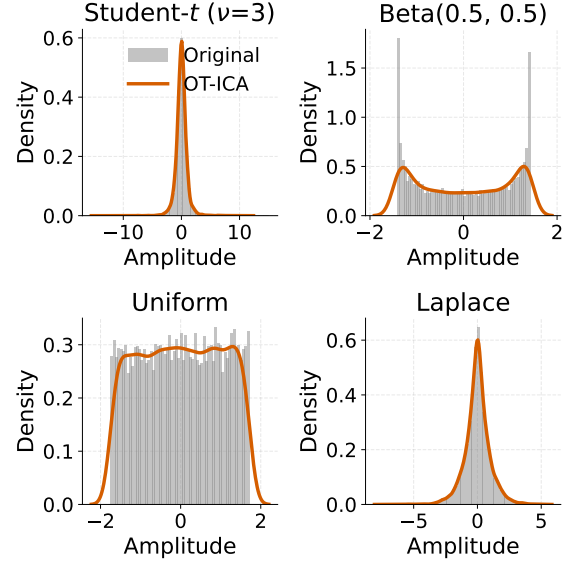


Figure 1: OT-ICA convergence on four standard continuous source types.  $W_2^2$  provides a smooth contrast that drives the Amari error to near zero across optimization iterations.

projecting gradients onto the tangent space of  $\mathcal{S}$  and retracting via symmetric decorrelation; and (iii) **Gaussian Dithering** [Schuchman, 1964], convolving discrete empirical distributions with a narrow Gaussian kernel to smooth step-function CDFs. Full formulations and pseudocode are in Appendix C.

## 3 OT-ICA METHODOLOGY VALIDATION

Separation quality is measured by the Amari Performance Index ( $E$ ) [Amari et al., 1996], the deviation of the gain matrix  $\mathbf{P} = \mathbf{B}_{\text{est}} \mathbf{A}$  from a generalized permutation matrix ( $E < 0.1$ : near-perfect;  $E \geq 0.5$ : failure; definition in Appendix D).

Figure 1 shows  $W_2^2$  driving the Amari error to near zero on Laplace, Uniform, Student- $t$ , and Beta sources, confirming that the Riemannian solver with analytical Gaussian targets converges reliably on continuous distributions.

## 4 EMPIRICAL ROBUSTNESS ON HYBRID MIXTURES

We compare OT-ICA against FastICA [Hyvärinen et al., 2001], JADE [Cardoso and Souloumiac, 1993], InfoMax [Bell and Sejnowski, 1995], and Picard [Ablin et al., 2018] on five mixture regimes at  $d \in \{10, 20, 30\}$ ,  $N = 10,000$ , 10 trials per condition (Figure 2).

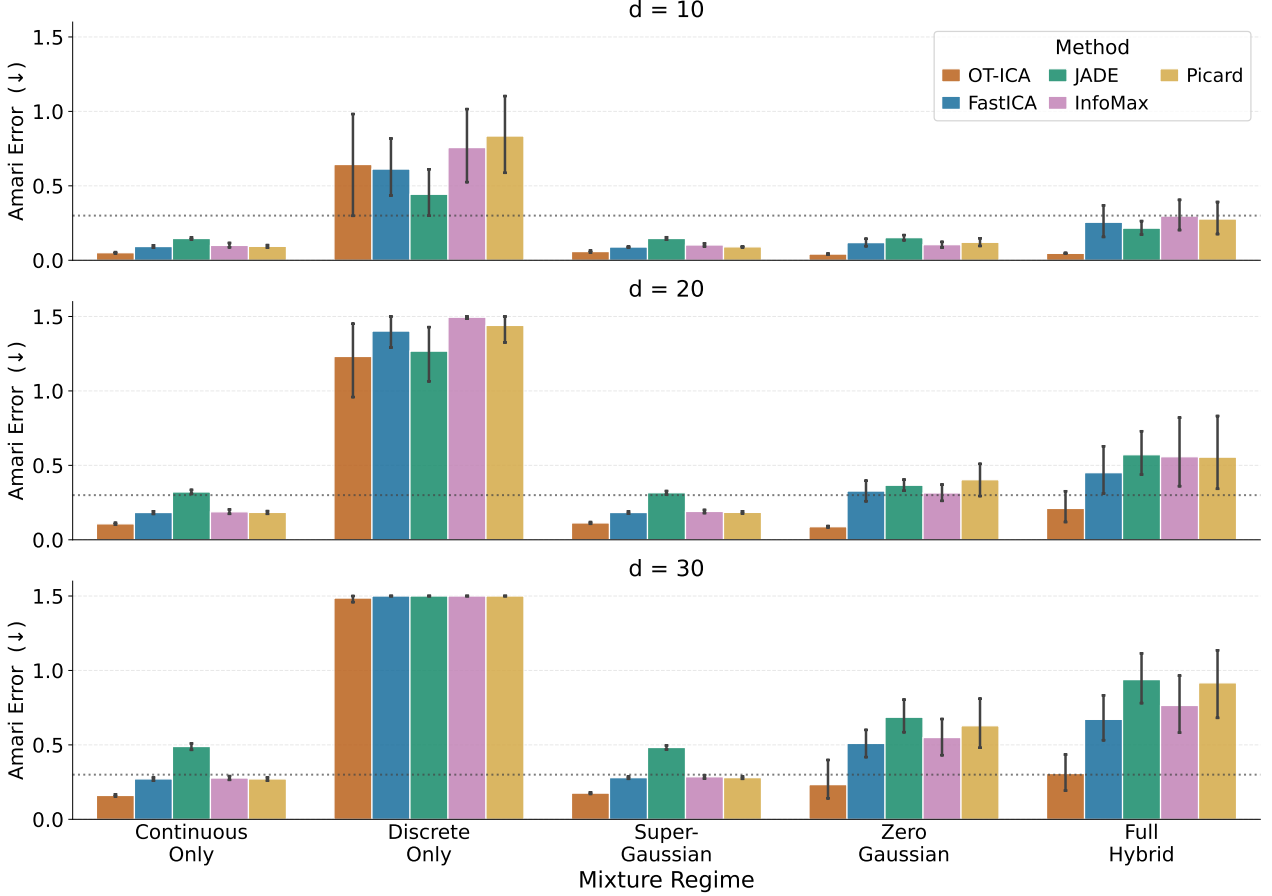


Figure 2: Mean Amari error ( $\downarrow$  better, clipped at 1.5) across five mixture regimes and  $d \in \{10, 20, 30\}$  ( $N = 10,000$ , 10 trials, 95% CI). OT-ICA achieves the lowest error on all continuous and heterogeneous regimes at every dimension. On *Discrete Only*, the  $W_2^2$  contrast is present but step-function CDFs produce gradient plateaus that stall the solver (Table 2, Appendix E.3). At  $d = 30$ , all proxy-based methods saturate the clip ceiling on discrete and several heterogeneous regimes.

**Continuous and heterogeneous regimes.** OT-ICA achieves the lowest Amari error on all four continuous and mixed-type configurations at every dimension. On *Continuous Only* sources, OT-ICA attains  $E = 0.051, 0.107, 0.160$  at  $d = 10, 20, 30$ , against the next best (FastICA: 0.092, 0.183, 0.271)—a reduction of 40–45% that holds across all three dimensions. The margin is larger on *Full Hybrid* mixtures (Laplace, Gaussian, Uniform, Student- $t$ , Beta combined): OT-ICA achieves  $E = 0.046, 0.211, 0.308$  against FastICA’s 0.255, 0.451, 0.671. On *Zero Gaussian* sources, where logcosh gradients degenerate for one component, OT-ICA yields  $E = 0.042, 0.088, 0.233$  while InfoMax yields 0.105, 0.315, 0.549.

**Discrete sources.** On *Discrete Only* mixtures, OT-ICA underperforms JADE at  $d = 10$  (0.643 vs. 0.443), is competitive at  $d = 20$  (1.231 vs. JADE 1.267), and all methods saturate the clip ceiling at  $d = 30$ . Table 2 shows  $W_2^2$  retains a clear non-Gaussianity signal on count data—exceeding logcosh resolution by more than  $10\times$  on standard Poisson( $\lambda = 3$ )—so the failure is not a contrast deficiency.

Step-function CDFs create gradient plateaus that stall the Riemannian solver despite the contrast being nonzero (Appendix E.3). Gaussian dithering (Appendix C) is the recommended mitigation for applications with discrete components, partially smoothing the CDF steps, but does not fully restore gradient signal at the scales tested.

**Method-specific failures.** FastICA and Picard [Ablin et al., 2018] produce nearly identical errors across all configurations (within 0.003 on continuous regimes at each dimension), establishing that Picard’s provably converging orthogonal solver does not close the gap: the bottleneck is the shared tanh nonlinearity, not the optimizer. JADE requires  $\mathcal{O}(d^4)$  samples for reliable fourth-order cumulant estimation—approximately 810,000 at  $d = 30$  against  $N = 10,000$  available—yielding  $E = 0.489$  on continuous and 0.938 on hybrid sources at  $d = 30$ . InfoMax [Bell and Sejnowski, 1995] is misspecified on distributions mixing super- and sub-Gaussian components, producing  $E = 0.765$  on hybrid at  $d = 30$  against OT-ICA’s 0.308. (Computational scaling in Appendix F.)

## 5 INTERDISCIPLINARY APPLICATIONS

### 5.1 STRUCTURAL IDENTIFICATION IN PRICE DISCOVERY

In fragmented financial markets, cointegrated prices share an unobserved common random walk. The Information Share (IS) [Hasbrouck, 1995, Baillie et al., 2002] quantifies each venue’s contribution, requiring decomposition of the reduced-form residuals  $u_t$  of a Vector Error Correction Model (VECM) [Engle and Granger, 1987, Johansen, 1991] into structural shocks  $\epsilon_t$  via  $u_t = \mathbf{B}\epsilon_t$ .

The covariance constraint  $\mathbf{\Omega} = \mathbf{B}\mathbf{B}^\top$  leaves  $\mathbf{B}$  unidentified up to an orthogonal rotation. Standard Cholesky decomposition resolves this by imposing a recursive market ordering, but IS estimates change with the ordering choice.

Factoring  $\mathbf{B} = \mathbf{S}\mathbf{C}$  (whitening times orthogonal rotation) reduces identification to finding  $\mathbf{C}$  [Zema and Cordoni, 2025]. OT-ICA recovers  $\mathbf{C}$  from the heavy-tailed non-Gaussianity of financial innovations without imposing a recursive ordering.

| Market / Source | True IS | Estimated IS (Mean) | Std. Dev. |
|-----------------|---------|---------------------|-----------|
| Market 1        | 0.1200  | 0.1212              | 0.0161    |
| Market 2        | 0.2400  | 0.2399              | 0.0245    |
| Market 3        | 0.6400  | 0.6389              | 0.0260    |

Table 1: Simulation Results for Information Share Recovery across 500 Monte Carlo runs. The estimated IS tightly matches the true IS without imposing arbitrary Cholesky recursive structures.

Table 1 reports mean IS estimates within 0.002 of the true values across 500 Monte Carlo runs, with standard deviations below 0.03, without imposing any market ordering.

### 5.2 EEG ARTIFACT REMOVAL

The skull acts as a linear volume conductor [Hyvärinen et al., 2001]: ocular blink artifacts mix with neural signals before reaching scalp electrodes. Blink artifacts are impulsive and super-Gaussian; their high  $W_2^2$  separates them from neural components without specifying a density model.

Figure 3 shows OT-ICA concentrating the blink artifact into a single component with excess kurtosis substantially above the remaining four. Zeroing that component and inverting the total unmixing matrix produces the RMS reduction quantified in the panel title, without tuning a density model.

## 6 CONCLUSION

OT-ICA replaces proxy contrast functions with the exact  $W_2^2$  metric computed by quantile sorting, avoiding FastICA’s

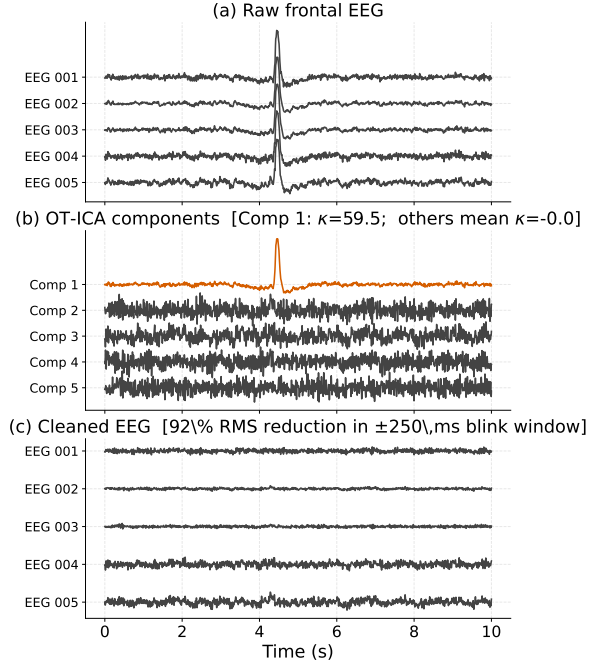


Figure 3: OT-ICA on five frontal EEG channels (MNE sample dataset). Panel (b): isolated blink component (orange) with excess kurtosis  $\kappa$  in the panel title; remaining components have mean  $\kappa$  near zero. Panel (c): signal reconstructed by zeroing the artifact component, with RMS reduction in the  $\pm 250$  ms blink window in the panel title.

gradient degeneracy, JADE’s  $\mathcal{O}(d^4)$  sample complexity, and InfoMax’s distributional misspecification. The Picard comparison isolates the contrast as the failure mode: a faster solver on the same tanh nonlinearity does not reduce Amari error. Across  $d \in \{10, 20, 30\}$  and five mixture regimes ( $N = 10,000$ , 10 trials), OT-ICA achieves the lowest Amari error on all continuous and heterogeneous configurations. On purely discrete sources, the contrast is present but gradient plateaus stall the solver—an open algorithmic problem. OT-ICA’s sorting-based contrast incurs  $\mathcal{O}(K \cdot dN \log N)$  per iteration against FastICA’s  $\mathcal{O}(dN)$ ; on purely continuous homogeneous sources, FastICA is faster. This tradeoff reverses at high dimensions on heterogeneous and hybrid mixtures, where proxy-based methods saturate the error ceiling while OT-ICA maintains separation (Appendix F). Applications to VECM structural identification and EEG artifact removal confirm the method transfers to real data without distributional assumptions. Future work will extend OT-ICA to causal discovery [Shimizu et al., 2006, Liang et al., 2023].

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# Supplementary Material for: Linear Independent Component Analysis via Optimal Transport Metric as Contrast

## A INFORMATION GEOMETRY OF ICA

Information geometry provides a framework for quantifying the relationship between independence and non-Gaussianity by analyzing probability distributions as points on geometric manifolds. Following the perspective introduced by Cardoso [2022], ICA can be understood as orthogonally projecting the empirical distribution  $P_Y$  onto two distinct subspaces: the Product manifold  $\mathcal{P}$  (containing distributions where marginals are strictly independent) and the Gaussian manifold  $\mathcal{G}$  (encompassing all multivariate Gaussians).

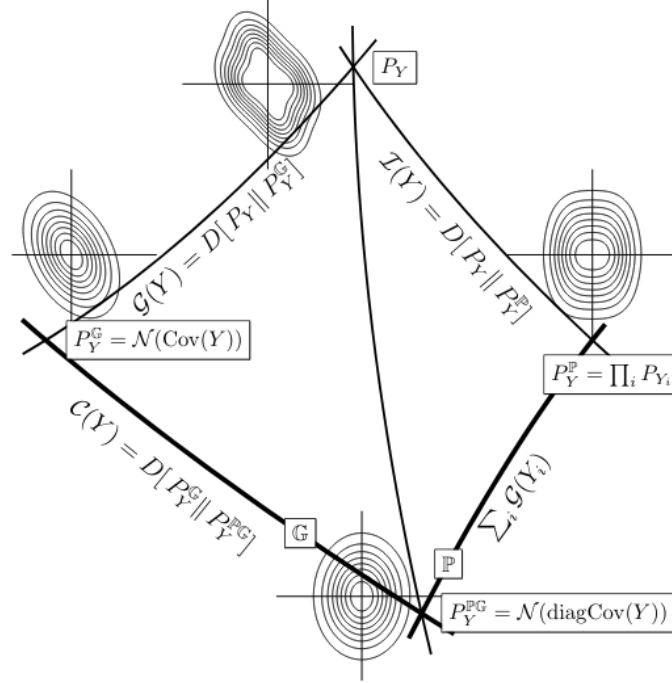


Figure 4: The Pythagorean geometry of Independent Component Analysis. Figure from Cardoso [2022].

The Mutual Information  $\mathcal{I}(Y)$  measures the distance from  $P_Y$  to its closest independent product distribution  $P_Y^P$ . A Pythagorean identity exists on the Product manifold for any candidate product distribution  $Q = \prod_i q_i$ :

$$\begin{aligned} D_{\text{KL}}(P_Y \| Q) &= \int P_Y(y) \log \left( \frac{P_Y(y) P_Y^P(y)}{P_Y^P(y) Q(y)} \right) dy \\ &= \mathcal{I}(Y) + \sum_i D_{\text{KL}}(P_{Y_i} \| q_i). \end{aligned} \quad (7)$$

A similar Pythagorean identity holds for the Gaussian manifold  $\mathcal{G}$ . Let  $\Phi_Y = \mathcal{N}(0, \text{Cov } Y)$  be the best Gaussian fit (sharing the exact mean and covariance of  $P_Y$ ), and let  $\Phi = \mathcal{N}(0, \Sigma)$  be an arbitrary Gaussian target. The divergence decomposes as:

$$\begin{aligned} D_{\text{KL}}(P_Y \| \Phi) &= \int P_Y(y) \log \left( \frac{P_Y(y) \Phi_Y(y)}{\Phi_Y(y) \Phi(y)} \right) dy \\ &= \int P_Y(y) \log \frac{P_Y(y)}{\Phi_Y(y)} dy + \int P_Y(y) \log \frac{\Phi_Y(y)}{\Phi(y)} dy \\ &= D_{\text{KL}}(P_Y \| \Phi_Y) + \mathbb{E}_{P_Y} [\log \Phi_Y(y) - \log \Phi(y)]. \end{aligned} \quad (8)$$

Because the log-density of a zero-mean Gaussian is a quadratic form, its expectation strictly depends only on the second moments (the covariance matrix) of the variable. Since the true distribution  $P_Y$  and its Gaussian fit  $\Phi_Y$  share the exact same covariance matrix by definition, evaluating the expectation of this quadratic form under  $P_Y$  is mathematically identical to evaluating it under  $\Phi_Y$ :

$$\mathbb{E}_{P_Y} \left[ \log \frac{\Phi_Y(y)}{\Phi(y)} \right] = \mathbb{E}_{\Phi_Y} \left[ \log \frac{\Phi_Y(y)}{\Phi(y)} \right] = D_{\text{KL}}(\Phi_Y \| \Phi). \quad (9)$$

Substituting this equivalence back establishes the Pythagorean identity on the Gaussian manifold:

$$D_{\text{KL}}(P_Y \| \mathcal{N}(\Sigma)) = D_{\text{KL}}(P_Y \| \mathcal{N}(\text{Cov } Y)) + D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\Sigma)). \quad (10)$$

We define the joint non-Gaussianity as  $G(Y) = D_{\text{KL}}(P_Y \| \mathcal{N}(\text{Cov } Y))$ , and correlation as  $C(Y) = D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\text{diag Cov } Y))$ . By evaluating the KL divergence from  $P_Y$  to a fully factorized Gaussian target  $\mathcal{N}(\text{diag Cov } Y)$  via both the Product path and Gaussian path, we yield Cardoso's unified theorem [Cardoso, 2022]:

$$\mathcal{J}(Y) + \sum_i G(Y_i) = G(Y) + C(Y). \quad (11)$$

Crucially, once the data is whitened ( $\text{Cov}(Y) = \mathbf{I}$ ), correlation drops strictly to zero ( $C(Y) = 0$ ). The relationship rearranges strictly as:

$$\mathcal{J}(Y) = G(Y) - \sum_i G(Y_i). \quad (12)$$

Because joint non-Gaussianity  $G(Y)$  is invariant under orthogonal rotation, minimizing mutual information (achieving independence) mathematically equates exactly to maximizing the sum of marginal non-Gaussianities  $\sum G(Y_i)$ .

## B MATHEMATICAL PROOFS FOR OT-ICA BOUNDS

### B.1 PROOF OF THEOREM 3: $W_2$ UPPER BOUND

**Theorem 5** ( $W_2$  Upper Bound). *Let  $\mathbf{Z} = (Z_1, \dots, Z_d)^\top$  be a vector of mutually independent latent sources, and let  $\mathbf{b}$  be a weight vector on the unit sphere ( $\sum_{i=1}^d b_i^2 = 1$ ). The squared Wasserstein distance between the linear mixture  $\mathbf{b} \cdot \mathbf{Z}$  and the standard normal distribution  $\Gamma$  is bounded by the weighted sum of the source distances to  $\Gamma$ :*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (13)$$

*Proof.* We construct a specific joint probability distribution, or coupling, using a common source of randomness. Let  $\mathbf{N} = (N_1, \dots, N_d)^\top$  be a vector of  $d$  independent standard normal variables,  $\mathbf{N} \sim \mathcal{N}(0, \mathbf{I}_d)$ , assumed to be statistically independent of the sources  $\mathbf{Z}$ . For each independent latent source  $Z_i$ , there exists an optimal transport map  $T_i$  that pushes forward the standard normal distribution to the source distribution, denoted as  $(T_i)_\# \Gamma = Z_i$ .

We construct a random variable  $X$  representing the mixture:

$$X = \sum_{i=1}^d b_i T_i(N_i). \quad (14)$$

To compare this mixture to a Gaussian distribution, we construct a target variable  $Y$  using the same underlying noise vector  $\mathbf{N}$ :

$$Y = \sum_{i=1}^d b_i N_i. \quad (15)$$

Given that the weights sum to unity, the resulting variable  $Y$  follows the standard normal distribution,  $Y \sim \Gamma$ . The pair  $(X, Y)$  creates a valid coupling. Because  $W_2^2$  is the infimum over all valid couplings, the optimal distance is less than or equal to the cost of our constructed coupling:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \mathbb{E} [|X - Y|^2] = \mathbb{E} \left[ \left( \sum_{i=1}^d b_i (T_i(N_i) - N_i) \right)^2 \right]. \quad (16)$$

Squaring the summation produces diagonal terms ( $i = j$ ) and cross terms ( $i \neq j$ ):

$$|X - Y|^2 = \sum_{i=1}^d b_i^2 (T_i(N_i) - N_i)^2 + \sum_{i \neq j} b_i b_j (T_i(N_i) - N_i)(T_j(N_j) - N_j). \quad (17)$$

Because the underlying noise variables  $N_i$  and  $N_j$  are independent, and the functions evaluate to a mean of zero, when we take the expectation, all cross terms vanish. We are left with the expectation of the diagonal terms:

$$\mathbb{E}[|X - Y|^2] = \sum_{i=1}^d b_i^2 \mathbb{E}[(T_i(N_i) - N_i)^2]. \quad (18)$$

Recognizing that  $\mathbb{E}[(T_i(N_i) - N_i)^2]$  is the definition of the squared Wasserstein distance between the individual source  $Z_i$  and  $\Gamma$ , we arrive at the bound:

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \quad (19)$$

## B.2 PROOF OF THEOREM 4: STRICT INEQUALITY

**Theorem 6** (Strict  $W_2$  Inequality for Non-Gaussian Sources). *Assuming at most one source  $Z_i$  is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3 is a strict inequality for any valid mixture where multiple weights  $b_i$  are non-zero:*

$$W_2^2(\mathbf{b} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^d b_i^2 W_2^2(Z_i, \Gamma). \quad (20)$$

*Proof.* In a one-dimensional setting, the  $W_2$  distance equals the expected squared difference of a specific coupling if and only if the random variables are comonotonic. By Brenier's Theorem [Brenier, 1991], this requires the gradients to be parallel at every point in the probability space:

$$\nabla X(\mathbf{N}) = \lambda(\mathbf{N}) \nabla Y(\mathbf{N}). \quad (21)$$

Assuming the source distributions possess smooth and strictly positive densities, Caffarelli's regularity theory [Caffarelli, 1992] guarantees that the optimal transport maps  $T_i$  are continuously differentiable.

We differentiate both sides to compute their Hessian matrices. On the LHS, the cross-derivatives  $\frac{\partial^2 X}{\partial N_i \partial N_j}$  are zero, yielding a diagonal matrix of  $b_i T_i''(N_i)$ . On the RHS, we differentiate the product  $\lambda(\mathbf{N})\mathbf{b}$ , yielding a Rank-1 outer product matrix:

$$\underbrace{\begin{bmatrix} b_1 T_1''(N_1) & 0 & \cdots & 0 \\ 0 & b_2 T_2''(N_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_d T_d''(N_d) \end{bmatrix}}_{\text{Hessian of } X \text{ (Diagonal)}} = \underbrace{\begin{bmatrix} b_1 \frac{\partial \lambda}{\partial N_1} & \cdots & b_1 \frac{\partial \lambda}{\partial N_d} \\ \vdots & \ddots & \vdots \\ b_d \frac{\partial \lambda}{\partial N_1} & \cdots & b_d \frac{\partial \lambda}{\partial N_d} \end{bmatrix}}_{\text{Outer Product } \mathbf{b}(\nabla \lambda)^\top \text{ (Rank-1)}} \quad (22)$$

Examining any off-diagonal entry ( $i \neq j$ ), the LHS dictates it must be zero, establishing  $b_i \frac{\partial \lambda}{\partial N_j} = 0$ . Because we assume a mixture where multiple weights are non-zero ( $b_i \neq 0$ ), it follows that the gradient of the scaling function is zero ( $\nabla \lambda = \mathbf{0}$ ). Consequently,  $\lambda(\mathbf{N})$  is a global scalar constant,  $\lambda$ .

Equating the diagonal elements implies  $T_i'(N_i) = \lambda$ . Integrating this indicates the optimal transport maps are linear functions of the form:

$$T_i(x) = \lambda x + c. \quad (23)$$

Applying a purely linear transformation to Gaussian noise  $N_i$  produces another Gaussian distribution. The identifiability assumption of ICA dictates that the original latent sources  $Z_i$  are non-Gaussian. Therefore, the linear map requirement contradicts the non-Gaussianity of the sources. Because the condition for equality cannot be met, the relationship is a strict inequality.  $\blacksquare$

### B.3 PROOF OF INTRACTABILITY OF NEWTON UPDATES FOR EMPIRICAL OT-ICA

**Theorem 7** (Intractability of Newton Updates). *For an empirical distribution mapped to a continuous target via sorting, computing a closed-form Hessian  $\mathbf{H} = \nabla_{\mathbf{b}}^2 W_2^2$  is analytically intractable without introducing external density estimation mechanisms. The Map Derivative  $\nabla_{\mathbf{b}} T_{\mathbf{b}}$  strictly requires the continuous Probability Density Function (PDF) evaluated at the quantile boundaries.*

*Proof.* To implement a fixed-point Newton step, the Hessian matrix of the squared Wasserstein distance,  $\mathbf{H} = \nabla_{\mathbf{b}}^2 W_2^2$ , is required. Let the 1D projection of the data be  $y = \mathbf{b}^\top \mathbf{X}$ , and let  $T_{\mathbf{b}}$  be the optimal transport map pushing this empirical projection to the target Gaussian distribution.

We compute the first differential (the gradient) with respect to  $\mathbf{b}$ . Utilizing the envelope theorem for optimal transport, the variation of the optimal map evaluates to zero within the cost function:

$$\mathbf{g} = \nabla_{\mathbf{b}} \left( \frac{1}{2} W_2^2 \right) = \mathbb{E} \left[ \mathbf{X} (\mathbf{b}^\top \mathbf{X} - T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})) \right]. \quad (24)$$

To compute the Hessian  $\mathbf{H}$ , we must differentiate the gradient vector  $\mathbf{g}$  with respect to  $\mathbf{b}$ :

$$\mathbf{H} = \nabla_{\mathbf{b}} \mathbf{g} = \mathbb{E} \left[ \mathbf{X} \mathbf{X}^\top - \mathbf{X} \nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})]^\top \right]. \quad (25)$$

Applying the total derivative using the chain rule yields two distinct components:

$$\nabla_{\mathbf{b}} [T_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X})] = \underbrace{T'_{\mathbf{b}}(\mathbf{b}^\top \mathbf{X}) \mathbf{X}}_{\text{Argument Derivative}} + \underbrace{(\nabla_{\mathbf{b}} T_{\mathbf{b}})(\mathbf{b}^\top \mathbf{X})}_{\text{Map Derivative}}. \quad (26)$$

By Caffarelli’s regularity theory [Caffarelli, 1992], the transport map  $T_{\mathbf{b}}$  is continuously differentiable, guaranteeing the existence of the Argument Derivative. The fundamental limitation lies within the Map Derivative.

In the 1D optimal transport setting, the transport map relies on the empirical cumulative distribution function,  $F_{\mathbf{b}}$ , of the projected data  $y$ . Expanding the Map Derivative yields:

$$(\nabla_{\mathbf{b}} T_{\mathbf{b}})(y) = \nabla_{\mathbf{b}} [\Phi^{-1}(F_{\mathbf{b}}(y))] = \frac{1}{\phi(\Phi^{-1}(F_{\mathbf{b}}(y)))} \nabla_{\mathbf{b}} F_{\mathbf{b}}(y). \quad (27)$$

The term  $\nabla_{\mathbf{b}} F_{\mathbf{b}}(y)$  represents the rate of change of the cumulative probability mass as the projection plane  $\mathbf{b}$  is rotated. The derivative of a cumulative distribution function’s boundary strictly requires the underlying continuous probability density function (PDF) evaluated at that specific boundary. For empirical distributions consisting of discrete sampled points, this density does not natively exist, rendering the closed-form calculation of the Hessian analytically intractable. ■

## C ALGORITHMIC ENHANCEMENTS AND METHODOLOGY

### C.1 WHY $W_2^2$ INSTEAD OF $W_1$ ?

While the  $W_1$  distance is robust to outliers, its reliance on an absolute  $L_1$  penalty introduces severe geometric challenges on finite samples.

As shown in Figure 5, the  $W_1$  cost lacks the convexity needed to smooth out empirical noise. Its derivative fluctuates frequently, causing first-order gradient ascent solvers to stall in local non-smooth traps. The squared penalty in the  $W_2^2$  metric acts as a structural smoother, penalizing local distortions and providing a continuous, decelerating gradient profile that allows for reliable convergence.

### C.2 ANALYTICAL GAUSSIAN TARGETS

Discrete approximation of the target Gaussian distribution introduces approximation noise. To eliminate this noise, we replace point-sampled targets with an analytical formulation, computing the expected value of a standard normal variable within each discrete quantile bin.

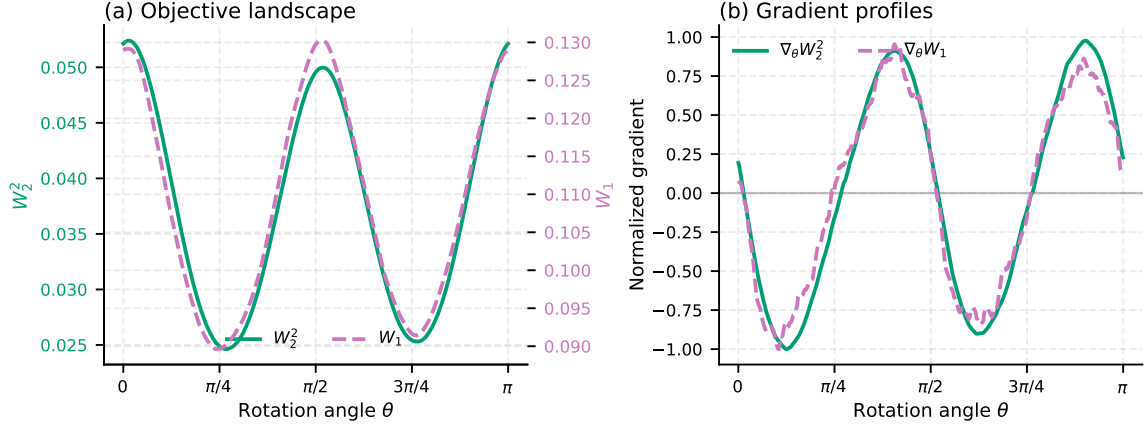


Figure 5: Gradient Landscapes of  $W_1$  and  $W_2^2$  across continuous rotation angles  $\theta$ . The  $W_2^2$  gradient smoothly decelerates toward zero, whereas the  $W_1$  gradient exhibits jagged volatility on finite empirical samples.

**Definition 1** (Analytical Gaussian Target). *Let the uniform probability bin edges for  $N$  samples be defined as  $p_i = \frac{i}{N}$  for  $i = 0, \dots, N$ , with corresponding Gaussian domain boundaries  $z_i = \Phi^{-1}(p_i)$ , where  $\Phi$  is the standard normal CDF. The analytical target value  $T_i$  for the  $i$ -th sorted sample is:*

$$T_i = N \int_{z_{i-1}}^{z_i} x \phi(x) dx = N(\phi(z_{i-1}) - \phi(z_i)). \quad (28)$$

### C.3 RIEMANNIAN GRADIENT AND RETRACTION

Because the input data is whitened, the unmixing matrix  $\mathbf{B}$  must remain in the Orthogonal Group  $\mathcal{S}$ , satisfying  $\mathbf{B}\mathbf{B}^\top = \mathbf{I}$ . Computing the standard Euclidean gradient  $\mathbf{G} = \nabla_{\mathbf{B}} W_2^2$  points into the unconstrained ambient space.

**Definition 2** (Riemannian Gradient on the Orthogonal Group). *Given the Euclidean gradient  $\mathbf{G}$ , the Riemannian gradient resulting from the projection onto the tangent space  $T_{\mathbf{B}}\mathcal{S}$  is given by [Absil et al., 2008]:*

$$\nabla_{\mathcal{S}} \mathbf{B} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top)\mathbf{B}. \quad (29)$$

**Definition 3** (Symmetric Decorrelation Retraction). *To map the matrix back to the orthogonal surface after a tangent step ( $\mathbf{B}_{\text{step}}$ ), we compute the overlap covariance  $\mathbf{C} = \mathbf{B}_{\text{step}}\mathbf{B}_{\text{step}}^\top$  and apply the inverse square root:*

$$\mathbf{B}_{\text{new}} = \mathbf{C}^{-1/2}\mathbf{B}_{\text{step}}. \quad (30)$$

### C.4 CONTINUOUS SMOOTHING VIA GAUSSIAN DITHERING

Applying continuous optimal transport directly to discrete mixtures introduces non-smooth step-functions. Adapted from signal processing [Schuchman, 1964], we inject a variance of continuous, zero-mean Gaussian noise ( $\sigma_{\text{dither}}^2 = 0.01$ ) into the 1D projected components prior to sorting. This decorrelates deterministic quantization errors and convolves the discrete empirical distribution with a Gaussian kernel.

### C.5 TOTAL COMPLEXITY AND STATISTICAL EFFICIENCY

For a dataset of dimension  $d$  with  $N$  samples,  $K$  random restarts (batched into a tensor  $\mathbf{B}_{\text{batch}}$ ), and  $T$  optimization iterations, the per-iteration complexity is:

$$\mathcal{O}(K \cdot (d \cdot N \log N + d^3)) \quad (31)$$

The  $N \log N$  term represents the sorting of projections, while the  $d^3$  term represents the cost of the symmetric decorrelation used for retraction.

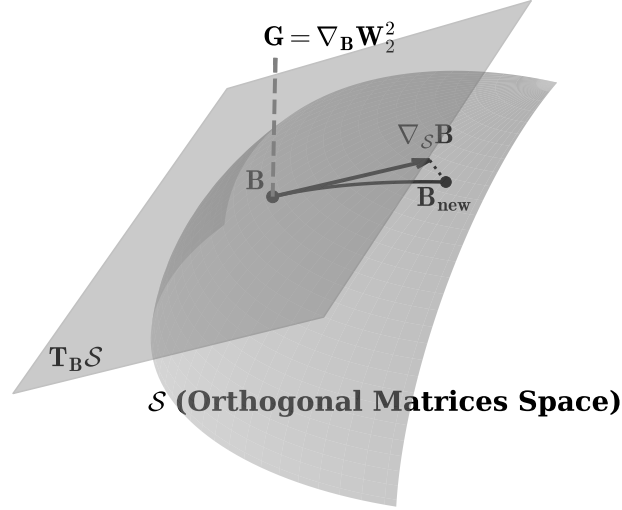


Figure 6: Geometric sketch of optimization on the Orthogonal Group  $\mathcal{S}$ . The Euclidean gradient  $\mathbf{G}$  is projected onto the tangent space to yield the Riemannian gradient  $\nabla_{\mathcal{S}} \mathbf{B}$ . A retraction maps the estimate back to the orthogonal surface.

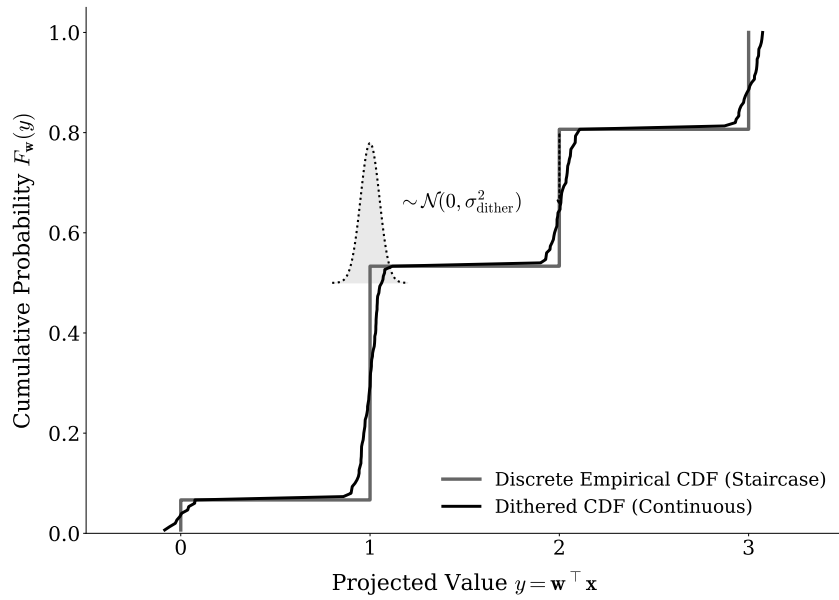


Figure 7: The non-differentiable staircase of a discrete empirical CDF (gray) is convolved with a narrow Gaussian kernel via dithering, yielding a continuous, differentiable curve (black).

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**Algorithm 1** The Optimal Transport ICA (OT-ICA) Algorithm

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**Require:** Observed mixture  $\mathbf{X} \in \mathbb{R}^{d \times N}$ , iterations  $K$ , learning rate  $\eta$ , batch size  $N_{\text{batch}}$ , dither noise  $\sigma_{\text{dither}}$

**Ensure:** Estimated unmixing matrix  $\mathbf{B}_{\text{final}} \in \mathbb{R}^{d \times d}$

```
1: Phase 1: Preprocessing & Target Generation
2: Center the data:  $\mathbf{X} \leftarrow \mathbf{X} - \mathbb{E}[\mathbf{X}]$ 
3: Whiten the data:  $\tilde{\mathbf{X}} \leftarrow (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X}$ 
4: Compute analytical target  $\mathbf{T} \in \mathbb{R}^{N_{\text{batch}}}$  via analytical Gaussian CDF integration
5: Phase 2: Initialization
6: Initialize  $\mathbf{B} \in \mathbb{R}^{d \times d}$  randomly from  $\mathcal{N}(0, 1)$ 
7: Apply symmetric decorrelation:  $\mathbf{B} \leftarrow (\mathbf{B}\mathbf{B}^\top)^{-1/2}\mathbf{B}$ 
8: Phase 3: Stochastic Riemannian Optimization
9: for  $k = 1$  to  $K$  do
10:   Sample stochastic mini-batch  $\tilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times N_{\text{batch}}}$  from  $\tilde{\mathbf{X}}$ 
11:   Project data onto candidates:  $\mathbf{Y} \leftarrow \mathbf{B}\tilde{\mathbf{X}}_{\text{batch}}$ 
12:   Apply Gaussian Dithering:  $\mathbf{Y}_{\text{dither}} \leftarrow \mathbf{Y} + \mathcal{N}(0, \sigma_{\text{dither}}^2)$ 
13:   Sort each row of  $\mathbf{Y}_{\text{dither}}$  ascending to yield empirical quantiles  $\mathbf{Y}_{\text{sorted}}$ 
14:   Compute Wasserstein objective:  $\mathcal{L} = \frac{1}{N_{\text{batch}}} \sum_{i=1}^d \|\mathbf{Y}_{\text{sorted}}^{(i)} - \mathbf{T}\|_2^2$ 
15:   Backpropagate to compute Euclidean gradient:  $\mathbf{G} = \nabla_{\mathbf{B}} \mathcal{L}$ 
16:   Project to tangent space:  $\nabla_{\mathcal{S}} \mathbf{B} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{B}^\top + \mathbf{B}\mathbf{G}^\top)\mathbf{B}$ 
17:   Update matrix along tangent plane:  $\mathbf{B}_{\text{step}} = \mathbf{B} + \eta \nabla_{\mathcal{S}} \mathbf{B}$ 
18:   Retract to Orthogonal Group  $\mathcal{S}$ :  $\mathbf{B} \leftarrow (\mathbf{B}_{\text{step}}\mathbf{B}_{\text{step}}^\top)^{-1/2}\mathbf{B}_{\text{step}}$ 
19: end for
20: Phase 4: Finalization
21:  $\mathbf{B}_{\text{final}} \leftarrow \mathbf{B}(\mathbf{X}\mathbf{X}^\top)^{-1/2}$ 
22: return  $\mathbf{B}_{\text{final}}$ 
```

---

## C.6 THE OT-ICA ALGORITHM

## D EXPERIMENTAL METHODOLOGY AND METRICS

### D.1 THE GENERALIZED PERMUTATION MATRIX AND AMARI INDEX

To empirically evaluate the success of source separation, we quantify the distance between the estimated unmixing matrix  $\mathbf{B}_{\text{est}}$  and the true mixing matrix  $\mathbf{A}$ . Because ICA is subject to scaling and permutation ambiguities, the target recovery is a generalized permutation matrix. Let  $\mathbf{P} = \mathbf{B}_{\text{est}}\mathbf{A}$  be the global transfer matrix. For a  $d \times d$  matrix  $\mathbf{P}$  with elements  $p_{ij}$ , the Amari Performance Index [Amari et al., 1996] measures the structural divergence:

$$E(\mathbf{P}) = \frac{1}{2d} \sum_{i=1}^d \left( \sum_{j=1}^d \frac{|p_{ij}|}{\max_k |p_{ik}|} - 1 \right) + \frac{1}{2d} \sum_{j=1}^d \left( \sum_{i=1}^d \frac{|p_{ij}|}{\max_k |p_{kj}|} - 1 \right). \quad (32)$$

### D.2 COMPUTATIONAL RESOURCE REGIMES

Because OT-ICA utilizes stochastic gradient descent while FastICA employs fixed-point Newton iterations, we standardize resources via two defined regimes to ensure fairness:

- **Low Compute Baseline:** OT-ICA is allocated restarts  $\min(4 \times d, 150)$  and 150/200 deflation/symmetric phase iterations. FastICA is allocated 50 random restarts and 1,000 maximum iterations per component.
- **High Compute Regime:** OT-ICA is allocated restarts  $\min(15 \times d, 600)$  and 300/500 iterations. FastICA is allocated 50 restarts and 5,000 maximum iterations per component.

## E LIMITATIONS OF PROXY CONTRAST FUNCTIONS

FastICA approximates negentropy using a non-quadratic contrast function  $G(\cdot)$ , such as the *logcosh* function. We identify two specific mathematical limitations of this approach.

### E.1 THE ZERO NEGENTROPY CONDITION

The approximation isolates non-Gaussianity via the difference of expectations:  $\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)]$ , where  $\nu \sim \mathcal{N}(0, 1)$ . If a latent independent source possesses a non-Gaussian distribution such that its expected value under the contrast function perfectly equals that of a Gaussian, the approximated negentropy evaluates to zero. The objective function provides no gradient signal, and the algorithm fails to extract the source.

### E.2 THE VANISHING CURVATURE CONDITION

FastICA utilizes a Newton (second-order) fixed-point iteration. Defining  $g(x) = G'(x)$  and  $g'(x) = G''(x)$ , FastICA approximates the Hessian matrix dynamically. For a given component  $i$ , the diagonal entry scales according to the expectation  $\mathbf{H}_{ii} \approx \mathbb{E}[s_i g(s_i) - g'(s_i)]$ .

The Newton update requires applying the inverse of this Hessian ( $\mathbf{H}^{-1}$ ). If a non-Gaussian source distribution causes the expectation in the denominator to evaluate to zero, the inverse becomes mathematically undefined. This analytical divide-by-zero error makes the update step intractable.

### E.3 DISCRETE DISTRIBUTIONS: CONTRAST VS. OPTIMIZATION

The *Discrete Only* failure is mechanistically distinct from the continuous failure modes above. Table 2 compares the non-Gaussianity resolution of  $W_2^2$  against the logcosh proxy negentropy  $(\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)])^2$  across standard and highly non-Gaussian parameterizations.

| Distribution                             | $W_2^2$ | Logcosh Negentropy |
|------------------------------------------|---------|--------------------|
| Laplace (continuous baseline)            | 0.0398  | 0.001214           |
| Binomial (standard, $n = 10$ )           | 0.0344  | 0.000031           |
| Poisson (standard, $\lambda = 3.0$ )     | 0.0513  | $\approx 0$        |
| Binomial (non-Gaussian, $n = 2$ )        | 0.2022  | 0.000267           |
| Poisson (non-Gaussian, $\lambda = 0.5$ ) | 0.3384  | 0.000085           |

Table 2:  $W_2^2$  provides  $\gg 10\times$  greater non-Gaussianity resolution than logcosh on count-based data. Standard Poisson ( $\lambda = 3.0$ ) evaluates to near-zero under logcosh yet retains a clear  $W_2^2$  signal. The failure to unmix discrete sources therefore cannot originate from an insufficient  $W_2^2$  contrast; the issue is in the optimization landscape.

### E.4 JADE: SAMPLE COMPLEXITY OF FOURTH-ORDER CUMULANTS

JADE [Cardoso and Souloumiac, 1993] identifies sources by jointly diagonalizing a set of fourth-order cumulant matrices via Jacobi sweeps. Reliable estimation of the full cumulant tensor requires  $\mathcal{O}(d^4)$  samples [Cardoso and Souloumiac, 1993, Hyvärinen et al., 2001]. At  $d = 30$ , this implies approximately 810,000 samples; against the  $N = 10,000$  available in our benchmark, the cumulant matrices are heavily undersampled. The resulting joint diagonalization converges on estimated tensors reflecting sampling noise rather than true source structure, yielding  $E = 0.489$  on continuous sources and  $E = 0.938$  on hybrid sources at  $d = 30$ —errors that approach or saturate the failure threshold despite JADE’s theoretical consistency on well-sampled data.

### E.5 INFOMAX: SCORE FUNCTION MISSPECIFICATION

InfoMax [Bell and Sejnowski, 1995] maximizes a log-likelihood under a fixed logistic nonlinearity, implicitly treating all sources as sub-Gaussian. On mixtures containing super-Gaussian components (Laplace, Student- $t$ ), the score function

### Discrete Mixture Optimization Landscape: OT-ICA vs. FastICA

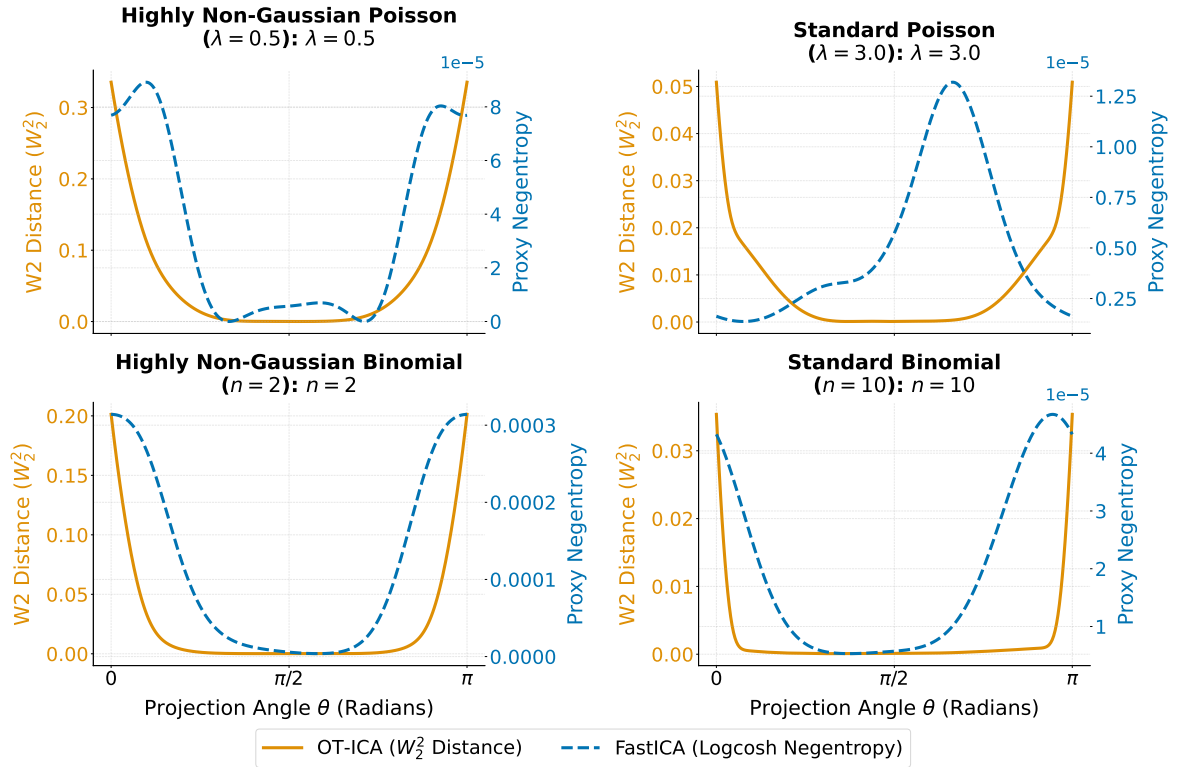


Figure 8:  $W_2^2$  and logcosh optimization landscapes for two-source Poisson and Binomial mixtures across parameterizations. The  $W_2^2$  landscape retains a non-zero signal throughout (confirming the contrast is present), but the step-function geometry of discrete CDFs creates flat plateaus where the gradient evaluates to near-zero, stalling gradient-based solvers. The logcosh landscape degenerates to near-zero for standard count-based parameterizations, indicating a fundamental contrast failure.

is misspecified: the gradient points away from the true unmixing direction. Lee et al. [1999] documented this failure mode explicitly, proposing extended InfoMax with online source-type switching. That approach partially addresses the misspecification but requires correct per-component classification—an assumption OT-ICA avoids entirely. In our benchmark, InfoMax produces  $E = 0.765$  on hybrid sources and  $E = 0.549$  on Zero Gaussian sources at  $d = 30$ , against OT-ICA’s 0.308 and 0.233 respectively, consistent with the misspecified-score explanation.

## E.6 PICARD: THE CONTRAST AS THE BINDING CONSTRAINT

Picard [Ablin et al., 2018] replaces InfoMax’s gradient ascent with an L-BFGS preconditioned solver on the same log-likelihood objective, achieving provably faster convergence per iteration. In our benchmark, FastICA and Picard produce Amari errors within 0.003 of each other across all continuous configurations at each dimension. This near-identical performance isolates the source of failure: solver speed is not the binding constraint. The shared tanh nonlinearity—and not the optimizer—determines the error floor. Replacing it with an exact, distribution-free contrast reduces error by 40–45% on continuous sources.

## F COMPUTATIONAL SCALING AND DIMENSIONALITY LIMITS

We benchmarked OT-ICA against FastICA using a linear mixture of continuous Laplace sources across increasing dimensions with a fixed sample size of  $N = 10,000$ .

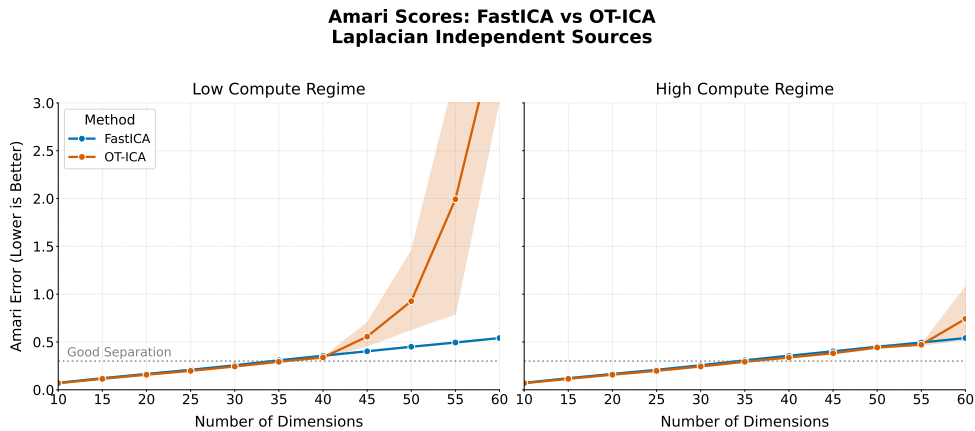


Figure 9: Amari error across dimensions for Laplacian sources at fixed  $N = 10,000$  samples. Both OT-ICA and FastICA hit the same curse-of-dimensionality noise ceiling at  $d = 40$ —an artifact of finite-sample sparsity shared by all empirical ICA methods at this regime, not a deficiency specific to OT-ICA.

While OT-ICA maintains error rates comparable to FastICA ( $E < 0.3$ ) in lower dimensions, both algorithms exhibit a loss in unmixing quality at  $d = 40$ . This limit is a manifestation of the curse of dimensionality. The expected error of the empirical Wasserstein distance scales as  $\mathcal{O}(N^{-1/d})$ . As  $d$  grows, a fixed finite sample fails to densely populate the state space, creating massive empty regions that swallow the gradient signal.

The execution time for OT-ICA scales more steeply than FastICA. FastICA evaluates to  $\mathcal{O}(dN)$  per iteration, making it computationally superior for exclusively homogeneous, continuous mixtures that do not trigger its proxy blind spots. OT-ICA incorporates parallel sorting across multiple restarts ( $\mathcal{O}(K \cdot dN \log N)$ ), trading execution speed for geometric robustness on complex topologies.

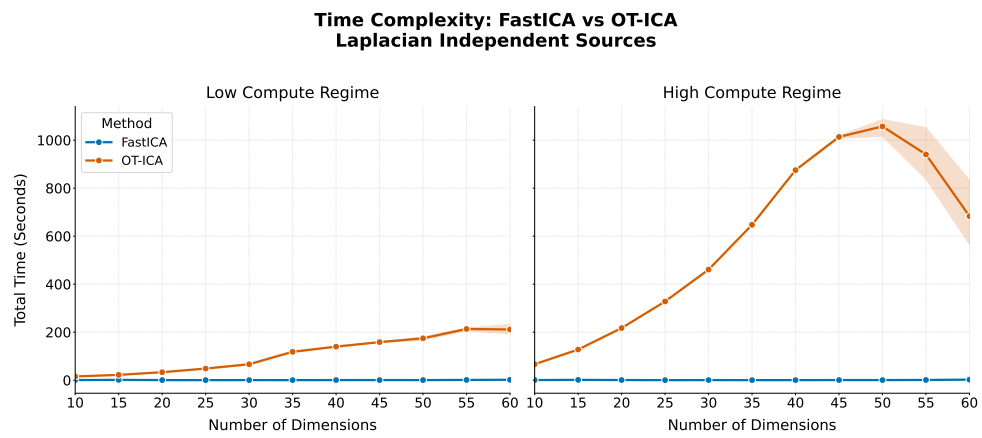


Figure 10: Total execution time (in seconds) for FastICA and OT-ICA across dimensions.